Venkata Lakshmi Parimala Pasupuleti

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PROFESSIONAL SUMMARY

AI Developer with 8+ years of production software engineering and hands-on agentic AI development. I build developer productivity platforms and production-grade agentic systems end-to-end: AgentProbe is a CI-integrated developer productivity platform -- 15-endpoint FastAPI backend, 5-metric LLM-as-judge eval engine, regression gate blocking failing agent runs in CI/CD, multi-provider LLM router (OpenAI/Anthropic/Groq/Ollama), observability dashboard, and golden dataset harness. Leasel is an 8-agent LangGraph pipeline with RAG, AI guardrails, human-in-the-loop escalation, and responsible AI controls (93.6% precision on 1,240 test cases). Strong hands-on experience with LangGraph, MCP, AWS (Bedrock, S3, Lambda, Glue), Node.js, TypeScript, Python, FastAPI, Docker, and CI/CD (GitHub Actions). Enterprise integration depth: REST APIs, internal API connectors, async pipelines (RabbitMQ), and third-party system integrations across 7+ years of production backend work. M.S. Data Analytics Engineering, GMU (GPA 3.87). AgentProbe: github.com/venkatalakshmiparimala/AgentProbe

CORE COMPETENCIES

Agentic AI & LangGraph | MCP Server Development | Developer Productivity Platforms | AWS Bedrock & AgentCore | CI/CD & SDLC Automation | AI Governance & Guardrails | Eval Frameworks & Observability | TypeScript, Node.js & Python

WORK EXPERIENCE

Blue Horse Digital

Sep 2026 -- Present

AI/ML Engineer (Volunteer) -- Virginia, USA

- Building agentic AI pipelines and ML models for business applications; integrating data systems and AI components with back-end APIs; iterating on production AI application behavior, observability, and reliability based on operational signals.

Magizhs Tech Solution

May 2025 -- Aug 2025

AI/ML Engineer Intern

- **Built and deployed a production agentic system using LangGraph** (parse → plan → retrieve → validate → conditional routing) -- integrated with enterprise data sources and internal APIs via REST; enforced AI guardrails blocking non-compliant outputs before delivery; engineered RAG pipeline (FAA Part 107 corpus) for grounded, cited compliance decisions.
- Implemented prompt engineering, function calling, and structured output validation; designed the full eval harness measuring solution effectiveness across 100+ test cases; deployed with Docker on AWS-compatible infrastructure.

George Mason University

Aug 2024 -- May 2026

Graduate Teaching Assistant + Research Associate

- Fine-tuned a custom classification model (HuggingFace Transformers + PyTorch); deployed inference API on Quart; designed model evaluation methodology and presented results to faculty stakeholders.
- Built multi-source prediction pipelines (Python, R, $R^2 = 0.977$); applied Isolation Forest for automated anomaly detection; designed Tableau/Quarto dashboards for operational reporting (+40% comprehension).

Entrata (Enterprise PropTech SaaS)

Jan 2017 -- Aug 2024

Senior Software Engineer -- Lehi, UT (Remote) [promoted from Associate, 2018]

- **Shipped 20+ production backend modules and enterprise integrations** across 120+ Agile sprints -- REST APIs, async pipelines (RabbitMQ), microservices (Docker, 50% scalability gain), third-party and internal API connectors for 100K+ monthly active users at 99.5% uptime; built internal developer tooling that reduced debugging time 40% and kept bug count below 10 for 6+ consecutive months.
- **Built and operated production data pipelines on AWS** -- S3 document migration with Libsodium encryption (100% PHI compliance, zero incidents), async processing with idempotency and retry guarantees, 180,000-record domain transformations from federal source data; optimized 200+ SQL queries (30% performance gain).
- Led SDLC automation initiatives eliminating 80% of manual operational workload through Python/Bash pipelines; drove technical standards via 300+ code reviews; mentored 5+ engineers on production engineering practices and observability.

PROJECTS

AgentProbe: Developer Productivity Platform for AI Agent Evaluation

FastAPI -- LangGraph -- CI/CD -- ChromaDB -- Observability

Production developer productivity platform for evaluating and governing LLM-powered applications: 15-endpoint FastAPI REST backend, 5-metric LLM-as-judge engine (correctness, groundedness, latency, token efficiency, hallucination rate), CI/CD regression gate automatically blocking failing agent runs before merge, multi-provider LLM router (OpenAI/Anthropic/Groq/Ollama), prompt version management, golden dataset harness, observability dashboard tracking solution performance over time. Cost controls and telemetry built in. Used to evaluate Leasel across 1,240+ compliance test cases with full audit trail.

github.com/venkatalakshmiparimala/AgentProbe

Leasel: Agentic AI Investigation Platform with Governance Controls

LangGraph 8-agent -- HITL -- FastAPI -- RAG -- AI Guardrails

Production agentic AI platform built on LangGraph: 8-agent stateful pipeline (Triage → Retrieval → Configuration → Historical → Structured Data → Reasoning → Validation → Human Escalation) over structured and unstructured data. RAG with hybrid search, citation grounding, and confidence scoring. Responsible AI controls: AI guardrails blocking unsupported outputs, HITL escalation when confidence too low, audit logging for every decision. FastAPI backend; CI/CD via GitHub Actions. Results: 93.6% precision / 87.9% F1 on 1,240 test cases; 100% citation correctness.

github.com/venkatalakshmiparimala/Lease_complaince_rag

AI on FHIR: NLP-Driven Clinical Records Query System

Flask -- LLMs -- BERT -- React -- Docker -- FHIR API

LLM-powered NLP platform translating natural-language queries (EN/ES) into FHIR API calls using LLMs, BERT, and embeddings; enterprise system integration with HIPAA-compliant RBAC; deployed with Docker; reduced manual clinical retrieval time ~60%. Demonstrates end-to-end AI integration with third-party APIs and enterprise data systems.

github.com/venkatalakshmiparimala/AI_on_FHIR

EDUCATION

M.S. Data Analytics Engineering -- George Mason University — May 2026

GPA 3.87 -- Virginia, USA

B.Tech Computer Science & Engineering -- JNTU Anantapur — 2017

SKILLS

AI & Agents: LangChain, LangGraph, Agentic AI, Multi-Agent Orchestration, MCP (Model Context Protocol), MCP Server Development, LLMs, Generative AI, RAG Pipelines, Prompt Engineering, Function Calling, Structured Outputs, AI Guardrails, Responsible AI, HITL, Human-in-the-Loop, Content Moderation, Prompt Injection Defense, Hallucination Detection, LLM-as-Judge, Eval Frameworks, Observability, Agent Tracing, OpenAI API, Anthropic API, Groq, Ollama, Claude Code, FastAPI, CI/CD (GitHub Actions)

Cloud & Infrastructure: AWS (Bedrock, AgentCore, S3, EC2, Lambda, Glue, Step Functions, SQS), Cloud-Native Application Development, Azure, GCP, Docker, Kubernetes, Terraform, Prometheus, Grafana, OpenTelemetry, RabbitMQ, Redis, PostgreSQL, Microservices, REST APIs, Enterprise API Integration, Internal APIs, Third-Party Integrations, Jira, GitHub, Confluence

Languages & Frameworks: Python, TypeScript, JavaScript, Node.js, SQL, FastAPI, Flask, React.js, Django, Java, Spring Boot

Developer Tooling & SDLC: Developer Productivity Platforms, CLI Tooling, SDLC Automation, CI/CD Pipelines, Custom Harnesses, Production Monitoring, Telemetry, Cost Management, Operational Controls, Regression Harnesses, Golden Datasets, Prompt Version Management, LLMOps, Audit Logging